

Mathematical Modelling — Tutorial 3

How these tutorials work. Each tutorial has pen-and-paper parts and a *computational task* that you carry out in whatever language you prefer (MATLAB is recommended; Python or Julia are equally fine). The point of the code is not the language, but that a classmate can regenerate your figure from your folder. Each week, you add a folder `weekNN/<username>/` to the course tutorials repository on GitHub, containing a half-page `report.md`, the figure(s) requested, the code that produced them, and a short `README` giving the command to regenerate them. You submit this as a *pull request* and review one classmate's in return (see the repository's `CONTRIBUTING.md`). Parts marked *Challenge* are optional, for the motivated or when time allows. Chapter and appendix references are to the course notes. Every `report.md` should also close with one sentence of *interpretation*: what does your result mean for the real system you modelled, and not merely as a piece of mathematics?

Tutorial 3 (Week 3 — Amplitude and period of a nonlinear oscillator)

Notes: Chapter 4.

For the undamped pendulum $\ddot{\theta} + \sin \theta = 0$ released from rest at amplitude θ_0 :

1. Using energy conservation, show that $\dot{\theta}^2 = 2(\cos \theta - \cos \theta_0)$, and hence that the period is the quadrature

$$T(\theta_0) = 4 \int_0^{\theta_0} \frac{d\theta}{\sqrt{2(\cos \theta - \cos \theta_0)}}.$$

2. Recall the Poincaré–Lindstedt prediction $T \approx 2\pi(1 + \frac{1}{16}\theta_0^2)$, and the small-amplitude (linear) value $T = 2\pi$.

Computational task. Evaluate the exact period $T(\theta_0)$ numerically for θ_0 from small up to near π (watch the integrable singularity at the upper limit). On one figure plot three curves: the exact $T(\theta_0)$, the linear value 2π , and the Poincaré–Lindstedt approximation. Identify the amplitude beyond which the perturbative curve visibly departs from the exact one — this is the function $\Phi(\theta_0)$ of Tutorial 1, now computed.

Repo deliverable. Submit:

- your code;
- the period-versus-amplitude figure;
- a `README.md` giving the command(s) to regenerate the figures, and
- a `report.md` commenting on the range of validity of the small-amplitude and Poincaré–Lindstedt models.